

Machine Learning as a Tool for Astrophysical Inference and Statistical Learning Foundations

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How to use these slides

Structure of Monday

- **Slot 1:** Machine learning as a tool for astrophysical inference.
- **Slot 2:** Statistical learning foundations.

Goal of the day

- Translate astrophysical questions into ML tasks.
- Understand the basic statistical learning setup.
- Recognize common failure modes before coding.

Guiding question

When does an ML prediction become astrophysical evidence?

Today's Agenda

Slot	Content
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|----|---|
| 1A | Motivation: why ML in astrophysics? |
| 1B | From scientific question to ML task |
| 1C | Data representations and astrophysical examples |
| 1D | Micro-discussion |
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|----|---|
| 2A | Statistical learning notation and risk minimization |
| 2B | Generalization, validation, and overfitting |
| 2C | Metrics, imbalance, leakage, and domain shift |
| 2D | Summary and exercise set orientation |
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Learning outcomes for Monday

By the end of this lesson, you should be able to:

- ➊ Identify whether an astrophysical problem is classification, regression, clustering, anomaly detection, representation learning, or generative modeling.
- ➋ Define features, labels, metadata, nuisance variables, and targets.
- ➌ Explain empirical risk minimization and the role of a loss function.
- ➍ Distinguish training error from generalization error.
- ➎ Choose metrics appropriate to imbalanced or rare-object problems.
- ➏ Detect common risks: leakage, shortcut learning, selection effects, and domain shift.
- ➐ Interpret ML outputs as uncertain, model-dependent scientific evidence.

What is changing in modern astrophysics?

Astronomy is increasingly driven by:

- large imaging surveys;
- high-cadence time-domain surveys;
- massive spectroscopic catalogs;
- multi-messenger alerts;
- numerical simulations and synthetic observations;
- multi-wavelength and multimodal datasets.

The bottleneck

Not only data volume, but also data complexity.

Human inspection alone cannot scale to millions or billions of objects.

Why ML? Three different motivations

Automation

Classify or filter massive numbers of objects.

- candidate selection
- alert triage
- catalog construction

Representation

Learn useful summaries of complex data.

- embeddings
- latent spaces
- feature discovery

Inference support

Approximate or accelerate scientific inference.

- emulators
- posterior estimation
- surrogate models

Important

ML is not a replacement for astrophysical reasoning. It is a statistical tool inside a scientific workflow.

Prediction is not the same as explanation

Predictive question

Given observables x , estimate a target y .

$$x \mapsto \hat{y}$$

Examples:

- classify a light curve;
- estimate redshift;
- rank transient candidates.

Scientific question

What physical process produced the observed data?

physics $\longrightarrow$ observations

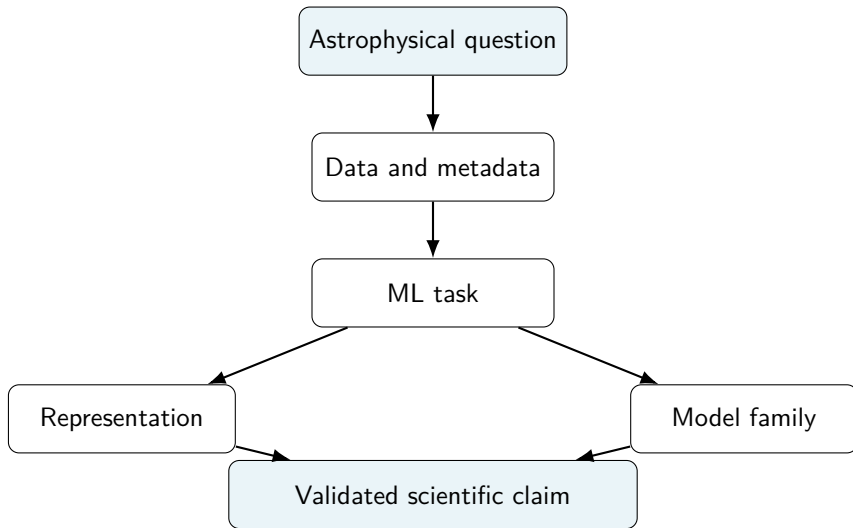
Examples:

- disk formation and dissipation;
- galaxy evolution;
- stellar pulsation modes.

Core warning

A model can predict well for the wrong physical reason.

Astrophysical question → ML formulation



A good ML project begins before the first model is trained.

The basic data object

Most supervised ML problems start from a dataset

$$\mathcal{D} = \{(\mathbf{x}_i, y_i)\}_{i=1}^N ,$$

where:

- $\mathbf{x}_i$ is the input representation;
- y_i is the target label or physical parameter;
- i indexes objects, events, spectra, images, or light curves.

In astrophysics, a more realistic object is often

$$(\mathbf{x}_i, y_i, \mathbf{m}_i, s_i, \sigma_i, c_i) ,$$

where $\mathbf{m}_i$ is metadata, s_i is survey/source information, σ_i encodes measurement uncertainty, and c_i may encode cluster, field, or environment.

Key idea

Metadata can be scientifically useful, but it can also create shortcuts.

Common astrophysical data geometries

Data type	Example observables	Natural models
Catalogs	magnitudes, colors, flags, morphology indices	trees, boosting, MLPs
Images	pixels, bands, cutouts, maps	CNNs, vision transformers
Spectra	flux as a function of wavelength	1D CNNs, transformers, PCA + regressors
Light curves	time, flux/magnitude, error, masks	features, RNNs, TCNs, transformers
Polarimetry	q, u, p, χ across bands/epochs	tabular ML, sequences, multimodal models
Simulations	fields, particles, parameters, snapshots	emulators, graph nets, CNNs

Representation principle

The model should respect the geometry of the data whenever possible.

A taxonomy of ML tasks in astrophysics

Supervised

- classification: class label $y \in \{1, \dots, K\}$;
- regression: continuous target $y \in \mathbb{R}$;
- ranking: prioritize candidates.

Unsupervised

- clustering;
- dimensionality reduction;
- anomaly detection.

Representation learning

- embeddings;
- self-supervised learning;
- transfer learning.

Generative / simulation-based

- synthetic observations;
- denoising;
- posterior estimation;
- emulation.

Example 1: photometric redshifts

Scientific problem

Estimate galaxy redshift when spectroscopy is unavailable.

ML formulation

$$(u, g, r, i, z, \dots) \longmapsto \hat{z}_{\text{phot}} .$$

- **task:** regression;
- **target:** spectroscopic redshift;
- **inputs:** magnitudes, colors, errors;
- **risk:** training set not representative of survey population.

Scientific caution

Good average error can hide catastrophic outliers in specific redshift or magnitude regimes.

Example 2: variable-star classification

Scientific problem

Identify classes of variable stars from light curves.

Possible representations

- engineered variability features;
- periodogram features;
- raw/resampled light curve;
- time-frequency representation;
- multimodal features + metadata.

Risks

- irregular sampling;
- cadence as a shortcut;
- class imbalance;
- survey-dependent labels;
- uncertain or ambiguous classes.

Lesson

The model choice is inseparable from the observing process.

Example 3: Be-star candidate identification

Astrophysical motivation

Classical Be stars are rapidly rotating B-type stars with circumstellar disks. Their photometric variability can contain signatures of pulsation, outbursts, and disk evolution.

ML task

light curve $\mapsto \mathbb{P}(\text{Be star})$.

- binary classification;
- candidate ranking;
- possible subtype classification.

Methodological issues

- strong class imbalance;
- heterogeneous surveys;
- label-resolution mismatch;
- source/domain shift;
- need for calibrated probabilities.

Example 4: polarimetry and disk geometry

Linear polarimetry gives normalized Stokes parameters

$$q = Q/I , \quad u = U/I .$$

From these, one can derive

$$p = \sqrt{q^2 + u^2} , \quad \chi = \frac{1}{2} \operatorname{atan2}(u, q) .$$

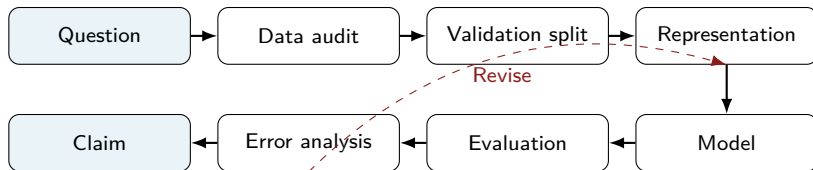
ML question

Can q, u, p, χ help infer disk inclination or viewing geometry?

Representation issue

Using χ directly may create artificial angular discontinuities. Sometimes (q, u) or $(p, \sin 2\chi, \cos 2\chi)$ is more natural.

The ML workflow in one slide



Scientific principle

Validation and error analysis are not afterthoughts. They decide what the result means.

What can go wrong? Shortcut learning

Shortcut learning

A model uses an easy correlate of the label instead of the intended physical signal.

Astrophysical examples

- learning survey ID instead of stellar class;
- learning image depth instead of morphology;
- learning S/N instead of spectral type;
- learning cadence instead of variability physics;
- learning color cuts instead of disk signatures.

Diagnostic

Train a model to predict the nuisance variable:

$$\mathbf{x} \mapsto \hat{s} \quad (\text{survey/source}) .$$

If this is easy, source shift must be handled explicitly.

Micro-discussion for Slot 1

Question 1

A classifier separates Be and non-Be light curves with high accuracy on a random mixed split. Later you discover that almost all Be labels came from Survey A and almost all non-Be labels came from Survey B. What did the classifier probably learn?

Question 2

For polarimetry, would you feed χ directly to a neural network? What could go wrong near the angular boundary?

Question 3

What would make a photometric-redshift model scientifically useful beyond a low global RMSE?

The supervised learning setup

Given training data

$$\mathcal{D}_{\text{train}} = \{(\mathbf{x}_i, y_i)\}_{i=1}^N ,$$

we choose a model family

$$f_{\theta} : \mathcal{X} \rightarrow \mathcal{Y} ,$$

where θ denotes model parameters.

Goal

Find parameters θ such that predictions generalize to new objects, not just the training sample.

Examples:

- $f_{\theta}(\text{image}) = \text{galaxy morphology}$;
- $f_{\theta}(\text{spectrum}) = T_{\text{eff}}$;
- $f_{\theta}(\text{light curve}) = \mathbb{P}(\text{variable class})$.

Loss functions

A loss function measures how wrong a prediction is:

$$\mathcal{L}(f_{\theta}(\mathbf{x}_i), y_i) .$$

Regression: mean squared error

$$\mathcal{L} = (y_i - f_{\theta}(\mathbf{x}_i))^2 .$$

Useful when large errors should be penalized strongly.

Binary classification: cross-entropy

$$\mathcal{L} = -y_i \log p_i - (1 - y_i) \log(1 - p_i) ,$$

where $p_i = f_{\theta}(\mathbf{x}_i)$ is a predicted probability.

Scientific point

The loss encodes what the training algorithm treats as important.

Empirical risk minimization

Training usually solves

$$\hat{\theta} = \arg \min_{\theta} \hat{R}(\theta) ,$$

where the empirical risk is

$$\hat{R}(\theta) = \frac{1}{N} \sum_{i=1}^N \mathcal{L}(f_{\theta}(\mathbf{x}_i), y_i) .$$

But the real quantity we care about is the expected risk

$$R(\theta) = \mathbb{E}_{(\mathbf{x}, y) \sim P_{\text{true}}} [\mathcal{L}(f_{\theta}(\mathbf{x}), y)] .$$

Central problem

We minimize risk on observed data, but we care about new data from the astrophysical population of interest.

Generalization gap

Training error

Error on the data used to fit the model.

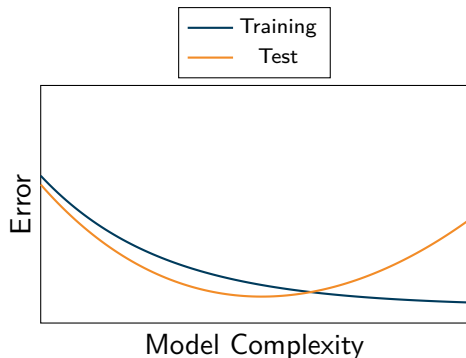
Test error

Error on independent held-out data.

Generalization gap

$\text{gap} = \text{test error} - \text{training error}.$

Large gaps suggest overfitting or distribution mismatch.



Underfitting and overfitting

	Underfitting	Overfitting
Cause	model too simple / poor representation	model too flexible for the data
Training error	high	very low
Validation/test error	high	much higher than training error
Astrophysical symptom	misses real structure	learns noise, artifacts, or survey shortcuts
Typical response	improve features/model	regularize, simplify, get more data, improve validation

Interpretation

The best model is not the one with the lowest training error, but the one that generalizes to the scientific use case.

Hypothesis space and inductive bias

The **hypothesis space** is the set of functions a model can represent:

$$\mathcal{H} = \{f_{\theta} : \theta \in \Theta\} .$$

Inductive bias

Every model makes assumptions about what patterns are easy to learn.

Model	Useful inductive bias
Linear model	simple global trends
Random forest / boosting	nonlinear tabular interactions
CNN	local spatial/translation structure
RNN/GRU/LSTM	sequential dependence
Transformer	long-range interactions via attention
Gaussian process	smoothness and uncertainty through kernels

Training, validation, and test sets

- **Training set:** used to fit parameters.
- **Validation set:** used to choose hyperparameters, architecture, threshold, stopping epoch.
- **Test set:** used once for final performance estimation.

Golden rule

The test set is not part of model development.



Splits must respect the physical unit of independence: star, galaxy, field, event, or simulation run.

Hyperparameters versus parameters

Parameters

Learned during training.

- neural network weights;
- regression coefficients;
- decision-tree splits.

Hyperparameters

Chosen by the user or validation procedure.

- learning rate;
- number of layers;
- tree depth;
- regularization strength;
- classification threshold.

Consequence

Using the test set to choose hyperparameters contaminates the final result.

Regularization

Regularization reduces overfitting by discouraging overly complex solutions.

Example: empirical risk with weight decay

$$\hat{\theta} = \arg \min_{\theta} \left\{ \frac{1}{N} \sum_{i=1}^N \mathcal{L}[f_{\theta}(\mathbf{x}_i), y_i] + \lambda \|\theta\|_2^2 \right\} .$$

Common regularization strategies:

- smaller model;
- weight decay;
- dropout;
- early stopping;
- data augmentation, if physically valid;
- better validation design.

Astrophysical note

Not all augmentations are physically valid. Do not augment away the signal you want to learn.

Preprocessing is part of the model

Examples:

- magnitude/flux conversion;
- standardization;
- continuum normalization;
- image resizing/cropping;
- light-curve resampling;
- missing-value imputation;
- outlier rejection.

Leakage risk

Fit preprocessing steps using only the training data.

For example, if you standardize using the mean and variance of the entire dataset, information from the test set leaks into training.

Rule

Save the preprocessing pipeline as carefully as the model.

Classification metrics: confusion matrix

For binary classification:

	Predicted positive	Predicted negative
True positive	TP	FN
True negative	FP	TN

$$\text{Precision} = \frac{TP}{TP + FP}$$

Of the predicted positives, how many are real?

$$\text{Recall} = \frac{TP}{TP + FN}$$

Of the real positives, how many did we recover?

$$F_1 = 2 \left(\frac{\text{precision} \cdot \text{recall}}{\text{precision} + \text{recall}} \right) .$$

Rare-object problem: why accuracy can fail

Suppose a catalog has 100,000 objects:

- 98,000 ordinary objects;
- 2,000 rare objects.

A trivial classifier predicts *ordinary* for every object.

$$\text{accuracy} = \frac{98,000}{100,000} = 98\%.$$

But

recall for rare objects = 0.

Astrophysical lesson

For rare-object discovery, high accuracy can be meaningless.

Balanced accuracy and MCC

Balanced accuracy:

$$BA = \frac{1}{2} \left(\frac{TP}{TP + FN} + \frac{TN}{TN + FP} \right) .$$

This averages recall for the positive and negative classes.

Matthews correlation coefficient:

$$MCC = \frac{TP \cdot TN - FP \cdot FN}{\sqrt{(TP + FP)(TP + FN)(TN + FP)(TN + FN)}} .$$

Use case

MCC is useful when classes are imbalanced because it uses all four entries of the confusion matrix.

ROC curves versus precision-recall curves

ROC curve

Plots true positive rate versus false positive rate.

$$\text{FPR} = \frac{FP}{FP + TN} .$$

Useful, but can look optimistic with strong imbalance.

Precision-recall curve

Plots precision versus recall as threshold changes. Especially informative when the positive class is rare.

Practical rule

For rare astrophysical objects, always inspect precision-recall behavior, not just ROC-AUC.

Regression metrics

For continuous targets, common metrics include:

$$\text{MAE} = \frac{1}{N} \sum_{i=1}^N |y_i - \hat{y}_i| ,$$

$$\text{RMSE} = \sqrt{\frac{1}{N} \sum_{i=1}^N (y_i - \hat{y}_i)^2} .$$

But also inspect residuals:

$$r_i = y_i - \hat{y}_i .$$

Astrophysical diagnostic

Plot residuals versus physical variables: magnitude, redshift, temperature, signal-to-noise, inclination, or survey field.

Cross-validation and grouped splits

Cross-validation estimates performance by repeating training/testing across multiple splits.

But in astrophysics...

Random cross-validation can be invalid if the same physical object appears more than once.

Examples requiring grouped splits:

- multiple spectra of the same star;
- several light-curve segments from the same object;
- repeated observations of the same galaxy;
- multiple simulated snapshots from the same parameter run;
- multiple events from the same host/source.

Rule

Split by the physical object, not by the row of the table.

Data leakage

Definition

Data leakage occurs when information unavailable at prediction time influences training, validation, or model selection.

Leakage type	Example
Duplicate leakage	same object in train and test
Preprocessing leakage	normalizing using all data before splitting
Target leakage	feature encodes the label or selection flag
Temporal leakage	using future observations to predict past/current state
Hyperparameter leakage	tuning repeatedly on the test set
Provenance leakage	source/instrument identity is accidentally aligned with labels

Leakage is not a small technical detail. It can invalidate the scientific claim.

Selection effects and domain shift

A model trained on one distribution may be applied to another:

$$P_{\text{train}}(\mathbf{x}, y) \neq P_{\text{test}}(\mathbf{x}, y) .$$

Sources of shift

- different survey depth;
- different cadence;
- different instruments;
- different sky fields;
- selection functions;
- label-definition changes.

Diagnostics

- per-source metrics;
- train-on-A test-on-B;
- source-classifier test;
- embedding visualization;
- calibration by domain.

Calibration: are probabilities meaningful?

A classifier may output $\hat{p} = 0.8$. What should that mean?

Calibration idea

Among objects assigned probability near 0.8, about 80% should truly belong to the positive class.

Mathematically, for a calibrated model:

$$\mathbb{P}(Y = 1 \mid \hat{p} \approx p) \approx p .$$

Why astrophysicists care

Candidate selection, follow-up prioritization, and catalog reliability often require trustworthy probabilities, not just class labels.

Uncertainty: two broad types

Aleatoric uncertainty

Irreducible uncertainty in the data-generating process.

- measurement noise;
- intrinsic stochastic variability;
- overlapping class distributions.

Epistemic uncertainty

Uncertainty due to limited knowledge or data.

- small training sample;
- poorly covered parameter regions;
- model uncertainty;
- domain shift.

Scientific interpretation

Uncertain predictions can be the most informative objects for follow-up.

Model comparison hierarchy

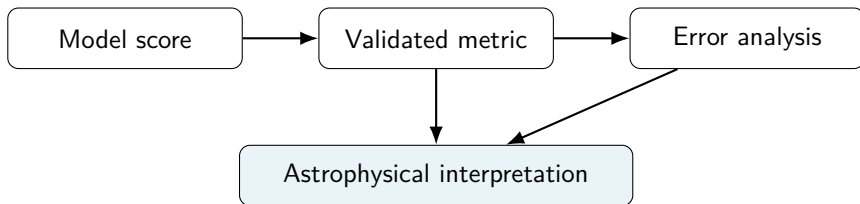
A responsible astrophysical ML study usually compares:

- ① trivial baseline: majority class, mean redshift, random ranking;
- ② simple interpretable baseline: linear/logistic regression;
- ③ strong classical baseline: random forest, gradient boosting;
- ④ specialized model: CNN, sequence model, transformer, multimodal model;
- ⑤ calibrated or uncertainty-aware final model.

Interpretation

A deep model is scientifically useful only if it improves, complements, or clarifies what simpler models already show.

From model output to astrophysical claim



Claim discipline

Do not claim “the model discovered the physics” unless the validation and error analysis support that statement.

Monday summary

- ① ML in astrophysics is a tool for prediction, representation, and inference support.
- ② A good ML project begins with the scientific question and data provenance.
- ③ Models learn correlations; astrophysical interpretation requires validation and error analysis.
- ④ Empirical risk minimization formalizes training, but generalization is the real goal.
- ⑤ Accuracy alone is often inadequate for imbalanced astrophysical tasks.
- ⑥ Leakage, selection effects, and domain shift can invalidate otherwise impressive metrics.
- ⑦ Calibrated probabilities and uncertainty estimates are essential for candidate selection and follow-up.

Connection to Exercise Set 1

Exercise Set 1 should be used to reinforce three levels of understanding:

Conceptual

Map astrophysical questions into ML tasks; identify risks and assumptions.

Mathematical

Use loss functions, metrics, risk, and validation notation correctly.

Scientific

Decide whether a model result supports an astrophysical claim.

Next lesson: neural networks and representation learning.

Optional reading and preparation

- Review basic probability: conditional probability, expectation, variance.
- Review linear algebra: vectors, matrices, dot products, norms.
- Review Python tools: NumPy arrays, pandas dataframes, matplotlib plots.
- Before Week 2, make sure you can run a Jupyter notebook with scikit-learn and PyTorch.

Preparation question

Bring one astrophysical problem from your own research and try to express it as:

input data $\longrightarrow$ target or scientific quantity .